

BIOL-1 Practice Test 01 — Answer Key

Modules 1-6: Study of Life through Metabolism

Part A: Multiple Choice Answers

Q	Answer	Explanation
1	C	All living things are made of cells
2	B	Observation → Hypothesis → Experiment → Conclusion
3	A	The independent variable is what the scientist changes
4	A	Atomic number = number of protons
5	B	Ionic bonds involve electron transfer
6	C	Water is polar due to unequal electron sharing
7	B	Buffers resist pH changes
8	D	Carbon forms the backbone of organic molecules
9	C	Monomers are building blocks that form polymers
10	D	Carbohydrates provide quick energy
11	D	Peptide bonds link amino acids
12	B	Cell theory: cells come from pre-existing cells
13	D	Prokaryotes lack a membrane-bound nucleus
14	A	Mitochondria produce ATP via cellular respiration
15	D	Plant cells have cell walls; animal cells do not
16	A	The fluid mosaic model describes membrane structure
17	B	Cell membrane = phospholipid bilayer
18	B	Phospholipids have hydrophilic heads and hydrophobic tails
19	D	Active transport requires ATP

Q	Answer	Explanation
20	A	Hypotonic = water moves INTO cell
21	B	Enzymes lower activation energy
22	C	ATP is the energy currency of cells
23	A	Substrates bind at the active site
24	D	Denaturation = loss of enzyme shape
25	D	Enzymes can be reused (not consumed)

Part B: Fill in the Blank Answers

Q	Answer
26	cell
27	nucleus
28	hydrolysis
29	acidic
30	amino acid
31	heterotrophs (or consumers)
32	endosymbiotic
33	osmosis
34	substrate
35	metabolism

Part C: Short Answer Key

36. Characteristics of living things (any 3):

- Cellular organization (made of cells)
- Metabolism (use energy, carry out chemical reactions)
- Homeostasis (maintain internal stability)
- Growth and development
- Reproduction
- Response to stimuli
- Adaptation/evolution

37. Four macromolecules:

- Carbohydrates, Lipids, Proteins, Nucleic Acids
- Protein monomer: amino acid

38. Passive vs. Active Transport:

- Passive transport: does not require energy (ATP); moves with concentration gradient (ex: diffusion, osmosis, facilitated diffusion)
- Active transport: requires energy (ATP); moves against concentration gradient (ex: sodium-potassium pump)

39. Enzyme Function & Specificity:

- Enzymes speed up reactions by lowering the activation energy required for the reaction to proceed.
- They are specific because the shape of their active site matches only specific substrate molecules (lock and key / induced fit).

End of Answer Key